

G:/code/vhdl/bus_multiplexer.vhd (/tb_bus_multiplexer/UUT) - Default

Ln#	
1	library ieee;
2	use ieee.std_logic_1164.all;
3	
4	entity bus_multiplexer is
5	Port(
6	A:in std_logic_vector(7 downto 0);
7	B:in std_logic_vector(7 downto 0);
8	C:in std_logic_vector(7 downto 0);
9	sel:in std_logic_vector(1 downto 0);
10	X:out std_logic_vector(7 downto 0)
11	);
12	end bus_multiplexer;
13	architecture Behavioral of bus_multiplexer is
14	begin
15	process(A,B,C,sel)
16	begin
17	case sel is
18	when "00" => X <= A;
19	when "01" => X <= B;
20	when "10" => X <= C;
21	when others => X <= "00000000";
22	end case;
23	end process;
24	end Behavioral;

Program 1: VHDL 8:1 Bus Multiplexer

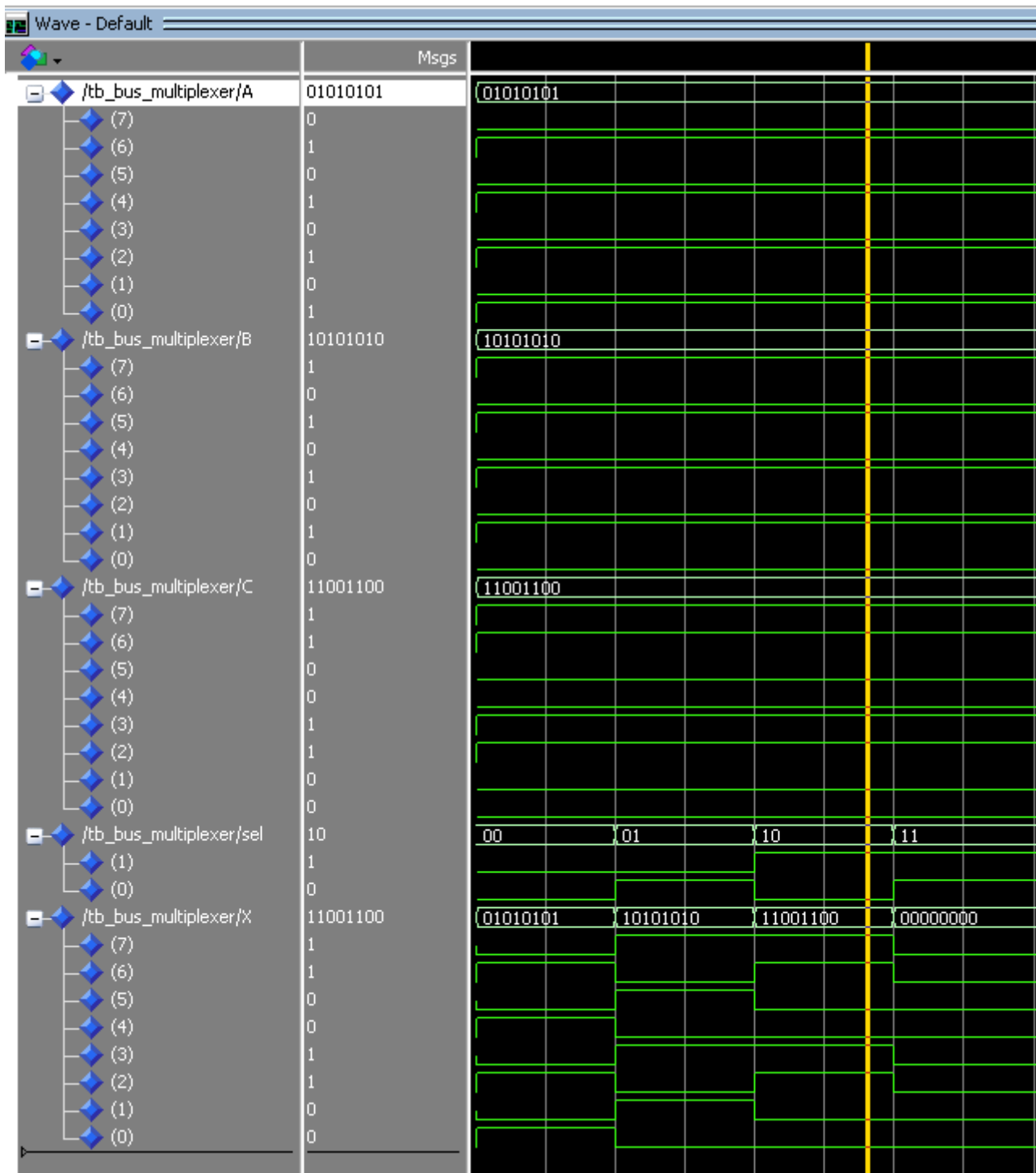
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G:/code/vhdl/tb_bus_muxiplexer.vhd (/tb_bus_muxiplexer) - Default
[Icons]

Ln#
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity tb_bus_muxiplexer is
5  end tb_bus_muxiplexer;
6
7  architecture Behavioral of tb_bus_muxiplexer is
8
9      component bus_muxiplexer
10     Port (
11         A      : in  STD_LOGIC_VECTOR(7 downto 0);
12         B      : in  STD_LOGIC_VECTOR(7 downto 0);
13         C      : in  STD_LOGIC_VECTOR(7 downto 0);
14         sel     : in  STD_LOGIC_VECTOR(1 downto 0);
15         X      : out STD_LOGIC_VECTOR(7 downto 0)
16     );
17 end component;
18
19 signal A      : STD_LOGIC_VECTOR(7 downto 0) := (others => '0');
20 signal B      : STD_LOGIC_VECTOR(7 downto 0) := (others => '0');
21 signal C      : STD_LOGIC_VECTOR(7 downto 0) := (others => '0');
22 signal sel     : STD_LOGIC_VECTOR(1 downto 0) := "00";
23 signal X      : STD_LOGIC_VECTOR(7 downto 0);
24
25 begin
26     A <= "01010101";
27     B <= "10101010";
28     C <= "11001100";
29     UUT: bus_muxiplexer
30     port map (
31         A      => A,
32         B      => B,
33         C      => C,
34         sel     => sel,
35         X      => X
36     );
37
38     stim_proc: process
39     begin
40
41         sel <= "00";
42         wait for 10 ns;
43
44         sel <= "01";
45         wait for 10 ns;
46
47         sel <= "10";
48         wait for 10 ns;
49
50         sel <= "11";
51         wait for 10 ns;
52
53         wait;
54
55     end process;
56
57 end Behavioral;
58

```

Program 2: VHDL Testbench for Bus Multiplexer



Testbench Output: Simulation Waveform of 8:1 Bus Multiplexer